Do Language Models Have Educational Personalities? Testing Recurring Behavior on New Tutoring Problems

Anonymous working draft

Incomplete working draft. Introduction, related work, methods, archive audit, measurement development, and limitations. Prospective confirmation results, the empirical abstract, discussion, and conclusion remain to be completed.

1 Introduction

Two language models can solve the same exercise correctly while teaching very differently. One gives the answer and explains it; another asks the student to complete a step; a third acknowledges the student's frustration before doing either. Such differences motivate a question about *educational personality*: do deployed models exhibit recurring teaching tendencies that help predict how they will respond in a new educational situation?

Answering this question requires more than distinguishing the models' text. Length, formatting, task instructions, and access to the solution can all produce recognizable differences. Nor does a model need to behave identically in every situation to have a recurring tendency. A tutor may consistently explain more after an erroneous attempt, or acknowledge difficulty specifically when a student expresses frustration. The empirical question is whether a model's defaults or its conditional response patterns recur sufficiently to predict new behavior. An observed difference, a repeatable measurement, and a successful prediction are distinct steps in that argument.

Education provides concrete choices through which to test this question without assuming a human personality taxonomy. Revealing a solution and inviting the student to reason are observable actions; they can co-occur, and neither is universally the better teaching move. Acknowledging expressed frustration is also observable without inferring that the model experiences empathy. These behaviors let us distinguish how a model tends to teach from how accurately it solves a problem or how much a learner benefits. Our object of study is the *tutor model's* recurring behavior. It differs from adapting instruction to an assigned or inferred *student personality*, as in personality-aware teaching systems (Rooein et al., 2026).

Prior studies identify model differences through personality-oriented scenarios and situated behavioral probes (Lee et al., 2025; Liu et al., 2026). Educational studies already compare characteristic tutoring-action mixtures and learn tutor-persona variation from dialogue (Kucheria et al., 2025; Lee et al., 2026). Building on these observations, we test the incremental predictive information in deployed models' defaults and student-state response patterns. The comparison is against the educational situation, model-specific subject habits, and simple training-output style profiles, with forecasts locked before new responses are generated.

We organize the investigation around three questions. **RQ1:** Under the same educational input, do models show distinguishable defaults that recur across repeated generations? **RQ2:** Can those defaults, or model-specific responses to student state, predict actions on independent source problems? **RQ3:** How do the resulting profiles change under neutral rewordings of the tutor role and explicit teaching instructions? These questions form one evidence chain. The first establishes recurring differences; the second tests their predictive value; the third identifies conditions under which they persist or move.

The study connects an existing educational-output archive to a separate prospective experiment. The archive contains more than one million raw prediction records, with different roles, coverage, and repetition histories. We audit those distinctions before using the archive for exploration. We then develop observable event measures on a small, permanently excluded pilot and freeze a source-separated training and confirmation design. Predictions are fitted and saved before confirmation responses are generated. Model-by-subject baselines prevent subject-specific habits from being mistaken for student-state response patterns, while neutral role wording and explicit teaching policies are manipulated separately on the same inputs.

The resulting claims concern observable tendencies of specified deployments. They do not establish human-equivalent traits, a latent psychological mechanism, or learning gains. In particular, a successful default profile does not imply that a more elaborate conditional profile will improve prediction, and a large instruction effect does not by itself establish or refute educational personality. The final conclusions must follow these comparisons, including measurement failures and limits on transfer.

2 Related Work

Personality and situated behavior. TRAIT extends personality inventories into scenario-based choices and evaluates distinctiveness, consistency, and prompting effects (Lee et al., 2025). Behavioral Mode Axes uses situated choices across interaction registers, also evaluates open-ended generations, and studies activation-based control (Liu et al., 2026). Separate construction and evaluation scenarios are already part of that framework. Internal-feature interventions further connect personality-related representations to behavior across situations (Zhang et al., 2026); conversational studies examine contextual variation under the same assigned personality (Han et al., 2026). Together these studies motivate examining both recurring defaults and conditional expression. Our deployment-level forecasts test observable educational actions and do not identify the internal mechanisms responsible for them.

Tutor behavior and educational personas. The CIMA comparison studies three LLMs in language-learning dialogues, reporting characteristic action mixtures and response complexity. Its generation prompt requests both a response and action labels (Kucheria et al., 2025). Thus, describing model-specific teaching styles is an existing research direction. Tutor-persona steering learns from human tutoring dialogues and evaluates persona alignment on held-out dialogues (Lee et al., 2026). Our experiment instead measures the default behavior of multiple deployed configurations and compares changes caused by controlled student cues and teaching instructions. We code the visible answers after generation and retain cross-coder sensitivity; this changes the measurement procedure without establishing that the resulting labels are ground truth.

Teaching quality and the predictive question. MRBench and MathTutorBench organize pedagogical-capability evaluation (Maurya et al., 2025; Macina et al., 2025). PATS studies teaching adapted to student personality (Rooein et al., 2026). Our primary target is the probability of the tutor's next observable behavior under specified conditions. Predicting that behavior does not rank its educational value. The empirical addition sought here is a joint account of which default and conditional profiles predict new-source actions, how those profiles compare with simpler explanations, and how neutral wording and explicit policies change them on matched inputs. Source separation and advance prediction locks make these comparisons auditable; their substantive contribution depends on the resulting evidence rather than on the procedure alone.

3 Study Design and Tests

We define *educational personality* operationally as recurring defaults and conditional response patterns that help predict a deployment's behavior on new educational inputs. The unit is a deployment configuration, including the requested and returned model names, provider, observation period, and generation settings. We study MiniMax-M3, MiniMax-M2.7, GLM-5.3, DeepSeek-V4-Pro, and Doubao-Seed-2.0-Lite. This definition neither assumes human traits nor requires an invariant ranking across contexts. Figure 1 shows how the design connects recurrence, prediction, and prompt effects. Complete materials, measurement rules, estimation details, and protocol history appear in Appendix B.

3.1 Exploration, Measurement Development, and Confirmation

The million-record archive informed exploration. Its 899,025 successful, variant-preserving deduplicated records span different roles; 68,799 belong to nine tutor-response settings under the current-adapter role audit. Historical prompt and deployment provenance is incomplete (Appendix A). We therefore keep archive findings separate from the prospective experiment. An excluded pilot of 160 answers develops the event measures reported in Section 4.

The prospective experiment contains 32 training questions in 30 conservative source families and 32 confirmation questions in 32 separate families. Each partition has eight questions in mathematics, science, language, and logic/data reasoning. Related training questions remain grouped during validation, and the four pilot templates and their numerical substitutions are excluded. Research agents authored the problems and student messages; two model reviewers checked all 64 question packages before freezing. The confirmation target is new sources within these four domains, not unseen domains or representative classroom interactions.

Each question crosses an erroneous attempt versus correct unfinished work with an explicitly calm versus frustrated student statement. The work contrast bundles correctness and progress; the affect contrast manipulates a textual cue. Every main condition supplies the same instructor reference solution, marked as unseen by the student. This equalizes solution availability without eliminating ability or instruction-following differences. Each model–input cell receives two requests at temperature 0.7 and an 8,192-token output budget. Training and canonical confirmation each contain 1,280 responses. Additional prompt arms contribute 2,560 responses on 16 confirmation sources. Secondary information-opportunity probes and cross-stage sentinels add 160 and 40 responses, respectively, for 5,320 planned tutor generations. These repetitions and variants do not add independent sources.

3.2 Observable Behavior and Recurrence (RQ1)

The three pilot-selected primary events are *answer revelation*, an explicit final resolution regardless of correctness; *reasoning elicitation*, an invitation for the student to perform reasoning that the tutor has not immediately answered; and *affect acknowledgement*, explicit recognition of feeling or difficulty beyond generic praise or politeness. These co-occurring actions are endpoints, not independent latent factors. Five secondary events remain reported regardless of their results.

MiniMax-M3, DeepSeek-V4-Pro, and GLM-5.3 each code every visible response using student context, the reference target, and original answer lines. Generator identity, role arm, repetition, and partition labels are withheld; output style may still reveal identity. Positive codes require

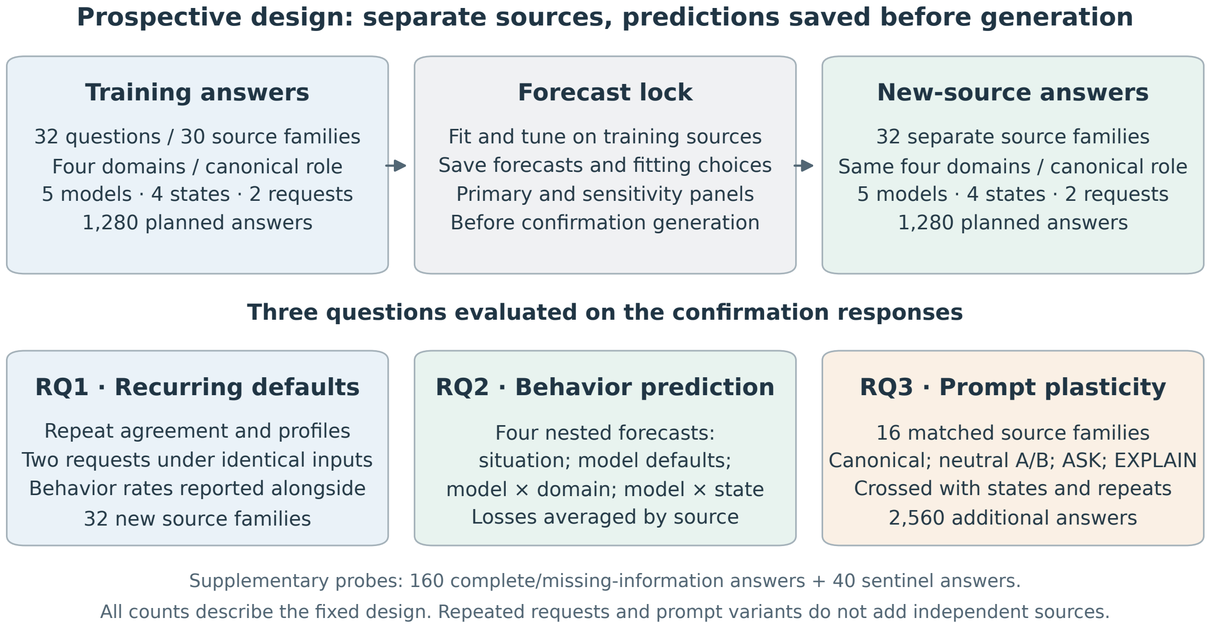

Figure 1: Prospective design, with planned response counts. Each main question crosses five configurations, four student-work/affect conditions, and two requests. Main training and confirmation source families are separate. RQ1 and RQ2 use the 32 canonical confirmation sources; RQ3 reuses 16 of them under additional role arms. Repetitions and prompt variants do not add independent sources. The four nested forecasts are accompanied by the training-only style competitor and the sensitivity analyses described below.

existing nonempty evidence lines. Primary labels use the majority of three structurally valid codes; traceability and agreement do not establish semantic ground truth. Repairs address completion or structure failures, never event labels or disagreement. One missing training code required a disclosed larger-budget amendment before prediction locking; Appendix B preserves its exact scope and the original failures.

RQ1 reports model event rates, state-specific profiles, and positive, negative, and total agreement across the two requests, with source clustering. Constant behavior can yield perfect agreement, so recurrence is interpreted alongside prevalence and predictive value.

3.3 Locked Prediction on New Sources (RQ2)

For each event, four nested logistic predictors successively include subject-by-student-state context, model identity, model-by-subject interactions, and model-by-student-state interactions. Training-only orthogonalization and standardization preserve the ridge geometry of earlier feature blocks. Source-grouped training validation selects the penalty from $\{0.1, 1, 10, 100\}$; the intercept is unpenalized. All fitted parameters, tuning choices, and confirmation probabilities are locked before any confirmation tutor response is generated. No confirmation answer or label enters prediction fitting. A descriptive style competitor uses only training profiles of log word count and marked-line fraction.

For each primary event, we test two Brier-loss improvements: adding model defaults to context, and adding student-state interactions to the model-by-subject predictor. Losses are averaged within source and equally across 32 confirmation sources. Six one-sided paired source-level tests use Holm correction; absolute losses, all gain signs, and source-bootstrap intervals are reported. Intervals condition on the fitted training profiles and a source-sampling approximation, not uncertainty over all educational settings or model families.

3.4 Prompt Effects and Alternative Explanations (RQ3)

On 16 sources selected by a frozen hash, each model receives every student state under the canonical role, two neutral role paraphrases, and explicit ASK and EXPLAIN instructions. Paired source-level event-rate differences measure both average displacement and model/state heterogeneity. The neutral equivalence reference is ± 0.10. Nominal multiplicity-adjusted paired-t intervals are retained, but the additional bounded-source check is too imprecise at sixteen sources to establish population equivalence. Nonsignificance does not establish stability. ASK/EXPLAIN effects quantify instruction-driven behavior, not better teaching. Joint revelation/elicitation rates allow both actions to co-occur without inferring their order.

Pre-confirmation sensitivity specifications include eight coding panels, five leave-one-generator analyses, and 200 training-source resamples. Mathematics and content-error screens evaluate the locked forecasts without refitting; flagged answers remove the complete five-model comparison slot. Two content reviewers also receive twelve agent-authored diagnostic controls. These checks assess dependence on measurement, configurations, training material, and errors; they do not establish human ground truth or causal control of ability. Their precise timing, retained coverage rules, and interpretation limits are given in Appendix B.

4 Measurement Development Results

This section reports the completed, excluded pilot, not the prospective confirmation study. Its 160 tutor responses cover only four templates and therefore do not justify education-wide prevalence or uncertainty estimates. All responses received three complete event codes after the recorded structural repairs. Figure 2 separates agreement on event presence from agreement on absence, rather than allowing abundant negative examples to dominate a single agreement statistic.

The three selected primary endpoints have majority-positive counts of 129 for answer revelation, 61 for reasoning elicitation, and 111 for affect acknowledgement. Their pairwise positive agreement ranges are 98.04–99.23%, 89.43–92.19%, and 97.32–98.63%, respectively. In contrast, requests for missing facts have only one majority-positive example: positive agreement ranges from 0–40% despite high overall agreement. Unsupported ability attribution also has weaker positive agreement, ranging from 33.33–51.16%. These results support different measurement priorities; they do not validate a fixed-dimensional personality taxonomy.

Pilot model profiles suggest potentially useful defaults, but adding student-state interactions does not consistently improve source-held-out prediction over model-by-subject profiles during method development. These exploratory observations motivated separate sources, within-domain replication, and locked prospective comparisons. They are not substituted for the missing confirmation outcomes.

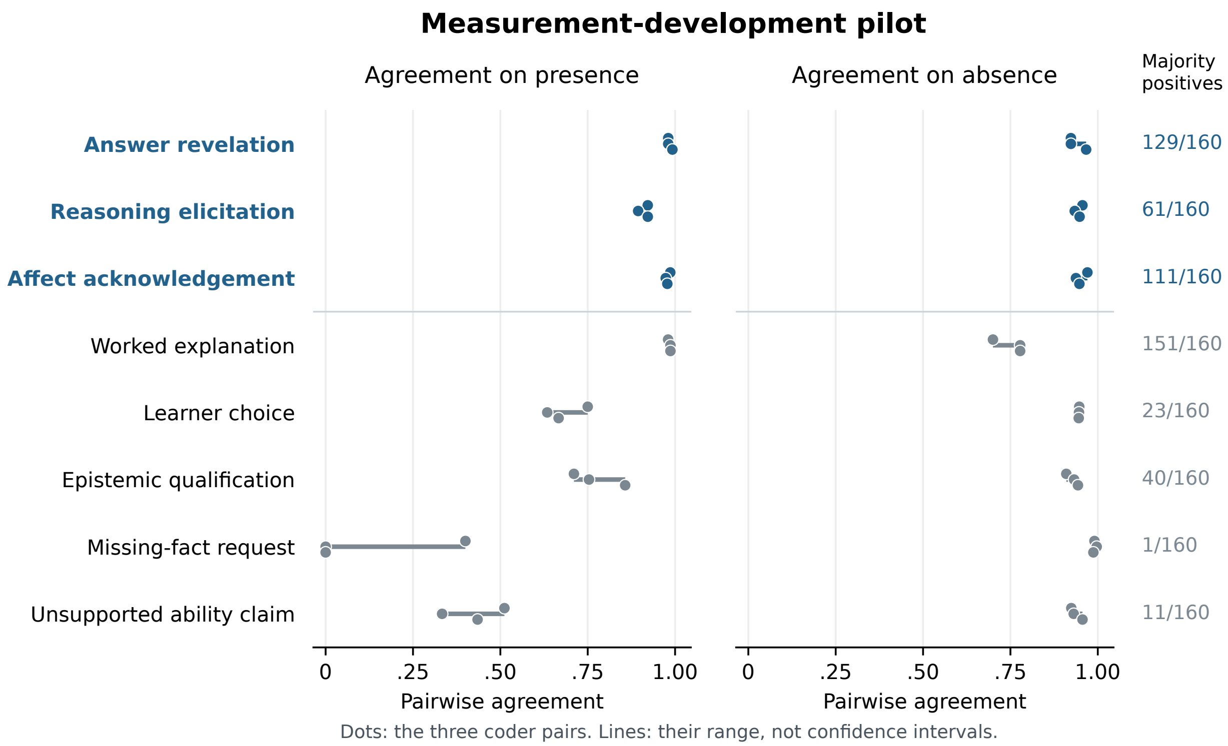

Figure 2: Completed measurement-development pilot: 160 tutor responses and 480 valid coder outputs. Each dot represents one of the three coder pairs; lines span the observed pairs and are not confidence intervals. Blue labels identify the three endpoints selected before formal generation. Agreement does not establish semantic ground truth, and rare events require separate attention.

5 Limitations

Scope of the construct. Educational personality denotes behavior observed in this educational setting. We do not compare matched educational and non-educational interactions, so the study cannot establish that a recurring tendency is unique to education. The three primary events cover a small set of actions, can co-occur, and do not constitute an independently validated latent trait structure. More predictable behavior also need not be more helpful: answer revelation includes incorrect conclusions, and elicitation can be appropriate or unhelpful depending on the learner's needs. No learning outcome is measured.

Materials and transfer. The prospective sources are agent-authored English problems with scripted student messages and supplied instructor references. They are not a probability sample of classroom interactions. Four domains and two explicit student-cue factors provide controlled comparisons, while leaving multilingual teaching, extended dialogue, implicit learner cues, and advanced subject matter outside the confirmation target. References equalize solution availability but cannot eliminate differences in understanding or instruction following. The source split prevents the specified problem families from crossing partitions; it does not establish novelty relative to model pretraining.

Measurement and deployment. Automated content review and three answer coders avoid new human annotation, but share possible model biases. Evidence-line checks verify location, and coder-family exclusions test a particular dependence; neither guarantees semantic validity. Rare or absent events require expression opportunities before their absence can be interpreted. Five selected deployments are also not a random sample of model families. Provider aliases and short-term repeated requests cannot demonstrate long-term stability of weights or behavior. The cross-stage sentinels mix generation variability, coding variability, and any deployment drift, so they cannot identify those causes separately.

Inference and research history. There are 32 confirmation source groups, not thousands of independent educational situations. Source intervals rely on an approximation to variation across problems resembling this constructed bank. Small gains may remain unresolved, and the conservative neutral-wording check cannot establish population equivalence with sixteen sources. Conditional prediction is limited to the selected student cues, feature blocks, and regularized model class; a weak incremental gain does not disprove all context-sensitive behavior. The archive analysis and pilot informed the study, and some sensitivity and interpretation rules were fixed during training. We disclose their timing and retain failed measurements. Local hashes and execution timestamps support an inspectable sequence, but are not independent third-party preregistration or tamper-proof timestamping.

References

Bin Han, Deuksin Kwon, and Jonathan Gratch. Personality expression across contexts: Linguistic and behavioral variation in LLM agents, 2026. URL https://arxiv.org/abs/2602.01063.

Wassily Hoeffding. Probability inequalities for sums of bounded random variables. *Journal of the American Statistical Association*, 58(301):13–30, 1963. doi: 10.1080/01621459.1963.10500830. URL https://doi.org/10.1080/01621459.1963.10500830.

Aayush Kucheria, Nitin Sawhney, and Arto Hellas. Comparing behavioral patterns of LLM and human tutors: A population-level analysis with the CIMA dataset. In *Proceedings of the 20th Workshop on Innovative Use of NLP for Building Educational Applications (BEA 2025)*, pages 873–881, 2025. doi: 10.18653/v1/2025.bea-1.64. URL https://aclanthology.org/2025.bea-1.64/.

Jaewook Lee, Alexander Scarlatos, Simon Woodhead, and Andrew Lan. Letting tutor personas speak up for LLMs: Learning steering vectors from dialogue via preference optimization. In *Proceedings of the 21st Workshop on Innovative Use of NLP for Building Educational Applications (BEA 2026)*, pages 78–92, 2026. doi: 10.18653/v1/2026.bea-1.7. URL https://aclanthology.org/2026.bea-1.7/.

Seungbeen Lee, Seungwon Lim, Seungju Han, Giyeong Oh, Hyungjoo Chae, Jiwan Chung, Minju Kim, Beong-woo Kwak, Yeonsoo Lee, Dongha Lee, Jinyoung Yeo, and Youngjae Yu. Do LLMs have distinct and consistent personality? TRAIT: Personality testset designed for LLMs with psychometrics. In *Findings of the Association for Computational Linguistics: NAACL 2025*, pages 8412–8452, 2025. doi: 10.18653/v1/2025.findings-naacl.469. URL https://aclanthology.org/2025.findings-naacl.469/.

Haoze Liu, Run Liu, Haiying Xu, Jiahui Han, Siyuan Fang, Siyu Yan, Huiqi Deng, Guanchu Wang, and Na Zou. Your LLM, your style: Behavioral mode axes for LLM behavioral control, 2026. URL https://arxiv.org/abs/2608.10703.

Jakub Macina, Nico Daheim, Ido Hakimi, Manu Kapur, Iryna Gurevych, and Mrinmaya Sachan. MathTutorBench: A benchmark for measuring open-ended pedagogical capabilities of LLM tutors. In *Proceedings of EMNLP 2025*, pages 204–221, 2025. doi: 10.18653/v1/2025.emnlp-main.11. URL https://aclanthology.org/2025.emnlp-main.11/.

Kaushal Kumar Maurya, Kv Aditya Srivatsa, Kseniia Petukhova, and Ekaterina Kochmar. Unifying AI tutor evaluation: An evaluation taxonomy for pedagogical ability assessment of LLM-powered AI tutors. In *Proceedings of NAACL-HLT 2025*, pages 1234–1251, 2025. doi: 10.18653/v1/2025.naacl-long.57. URL https://aclanthology.org/2025.naacl-long.57/.

Donya Rooein, Sankalan Pal Chowdhury, Mariia Eremeeva, Yuan Qin, Debora Nozza, Mrinmaya Sachan, and Dirk Hovy. PATS: Personality-aware teaching strategies with large language model tutors. In *Findings of the Association for Computational Linguistics: EACL 2026*, pages 4186–4211, 2026. URL https://aclanthology.org/2026.findings-eacl.219/.

Ruikang Zhang, Shuo Wang, and Qi Su. From representations to behaviors: Exploring the person–situation–behavior triad in LLMs, 2026. URL https://arxiv.org/abs/2607.26853.

A Archive Scope and Historical Provenance

The frozen archive census contains 1,028,822 raw records. Successful text deduplication yields 899,025 model–benchmark–item–response records when recorded input variants remain distinct, and 837,704 when variants are ignored for text inventory only. Neither count estimates independent generation events. Table 1 classifies all 39 benchmark settings using a review of current adapter contracts. Each assignment retains a source file, class, line, hash, and interpretation note. Current contracts guide role interpretation but do not establish the exact prompts used by every historical request.

Current task role	Settings	Records
Evaluation of existing work	7	354,308
Test or option answering	10	306,307
Mixed educational assistance	2	78,322
Tutor response	9	68,799
Learner history or diagnosis	2	42,997
Safety response	2	13,974
Metacognitive problem solving	3	11,188
Constrained question generation	1	9,233
Assigned solution repair	1	8,015
General instruction following	1	3,782
Knowledge-structure reasoning	1	2,100
Total	39	899,025

Table 1: Variant-preserving text inventory, not counts of independent tutoring contexts or personality labels. Mixed educational assistance requires item-level role separation; it is not automatically excluded from later research.

Evaluation and test/option outputs comprise 73.48% of these records. For example, the Pedagogy Benchmark requests pedagogical-knowledge multiple-choice answers, while ASAP and SAS ask the candidate model to score student work. MRBench and BEA judge settings likewise evaluate existing tutor text; that text is not the candidate evaluator's own tutoring. Even among tutor settings, instructions differ: TutorBench use cases request different actions, MMTutorBench requires image-conditioned guidance in three prescribed sections, and LongTutor explicitly requests scaffolding without the full answer.

A second pass verifies all 122 prediction files and corresponding summaries in the nine tutor-response settings against the census hashes. Their 86,072 latest successful file items reproduce 68,799 variant-preserving deduplicated records. None of those file items contains the inspected fields for complete messages, prompt text or input hashes, provider request identifiers, returned model names, generation timestamps, or finish reasons. All contain usage metadata. Of the 122 summaries, 83 contain generation parameters and 114 contain a judge-prompt hash; none contains the inspected generation-prompt hash fields. A judge-prompt hash is not evidence of the tutor's generation prompt, and later scoring summaries do not necessarily attest to the original generation parameters.

This is a field-coverage finding, not proof that provenance is unavailable in external logs or original sources. Reconstructed historical inputs must be identified as reconstructions. Consequently, the archive provides discovery and confound audits, while the prospective study records the input, output, source, version, and timing relationships needed for its own narrower confirmation.

B Detailed Methods and Protocol History

We use *educational personality* operationally for recurring default behavior and conditional response patterns of a deployed model in educational interaction. Let $Y^{(k)}_{mqsar}$ denote whether model configuration m expresses behavior k on source problem q, student state s, instruction arm a, and independent request r. A default profile summarizes behavior under the canonical tutor role over a specified distribution of educational inputs. A conditional profile additionally represents how that behavior varies with student state. We test both descriptions through predictions on separate source problems. This operationalization does not require a fixed cross-context model ranking, and it does not treat any identifiable linguistic signature as a validated personality construct.

The unit m is a deployment configuration, including its requested and returned model names, provider, observation period, and generation settings. The five prospective configurations request MiniMax-M3, MiniMax-M2.7, GLM-5.3, DeepSeek-V4-Pro, and Doubao-Seed-2.0-Lite. In the measurement pilot, the request alias GLM-5.2 returned GLM-5.3 throughout; those responses are identified by the returned name and are not pooled with historical GLM-5.2 results. Provider model names do not attest to immutable weights, so version and temporal uncertainty remain limitations of deployment-level inference.

B.1 Archive Exploration and Prospective Separation

The archive audit distinguishes raw records from successful deduplicated responses, and distinguishes tutors from solvers and evaluators. Repeated questions and conversations can occur under different benchmark names and prompt variants. Consequently, benchmark names and local item identifiers are not sufficient independence units. Related source material must stay together when estimating transfer. Analyses informed by previous archive results remain exploratory; starting a new conversational context does not make them prospective.

Across the 899,025 variant-preserving text records, a current-adapter role review identifies 68,799 records in nine tutor-response settings; most other records come from evaluation or test-answer tasks. Tutor-role membership does not itself establish neutral prompts or equivalent historical requests. Appendix A reports the complete role inventory and a hash-verified inspection of the 122 tutor prediction files, including their missing generation-provenance fields.

The controlled protocol uses 32 training questions in 30 conservative source families and 32 confirmation questions in 32 separate source families. Each partition has eight questions in each of mathematics, science, language, and logic/data reasoning. Two training question pairs share an averaging or unit- conversion family and remain grouped in internal validation. The four pilot templates and their numerical substitutions are excluded. Confirmation tests new sources within these four domains, rather than transfer to entirely new academic domains or proof that a model has never encountered the underlying concepts.

The questions, correct references, erroneous student attempts, and correct unfinished work were authored by the research agent. Two model reviewers checked 64 question packages before freezing, yielding 128 complete reviews. All five Boolean content checks were positive in every review. Twenty-five reviews nevertheless included issue text, mostly describing the deliberately incorrect attempt or the deliberately unfinished work; these comments and their agent adjudication are retained. This is content quality control without new human annotation, not a human-gold validation claim.

B.2 Crossed Educational Inputs

Each main question has two student-work states, an erroneous attempt and correct but unfinished work, crossed with two explicit affect statements, calm and frustrated. These are four controlled input conditions. The work-state contrast bundles correctness and progress; it does not isolate a pure progress or latent student-ability effect. The affect statement is an input manipulation, not an observation of a real learner's emotion.

All prospective main conditions provide the tutor with the same correct instructor reference for that question, specifying that the student has not seen it. This equalizes the availability of the solution but does not fully control comprehension, attention, or instruction-following ability. Results under this controlled reference condition are reported separately from the historical archive.

The training stage uses the canonical tutor role on all 32 questions. The confirmation stage first repeats this design on its 32 new questions. Within each confirmation domain, four questions are selected by a fixed hash for four additional role arms: two neutral paraphrases, an instruction to prioritize student reasoning without revealing the final answer, and an instruction to explain the solution and state the answer. Each selected question receives every arm and student state. The neutral arms change only the role wording; the two policy arms explicitly change the requested teaching behavior.

Every model–input cell receives two requests at temperature 0.7 with an 8,192-token output budget. The fixed design specifies 1,280 training responses, 1,280 canonical confirmation responses, and 2,560 additional intervention responses. It also includes 160 secondary responses to paired complete/missing- information probes on eight confirmation sources, plus 40 cross-stage responses to four training-source sentinels. Sentinels and prompt variants do not increase the count of independent confirmation sources. Request manifests, input hashes, parameters, timestamps, and failure histories are recorded. Requests are interleaved within shared provider concurrency limits rather than run as separate model blocks.

B.3 Observable Events and Measurement Development

Three primary events are fixed from the measurement-development pilot. *Answer revelation* records an explicit final resolution, including an incorrect resolution or a correct statement that information is insufficient. It therefore measures disclosure, not correctness. *Reasoning elicitation* records an invitation for the student to perform unanswered reasoning, including imperatives and indirect suggestions; rhetorical questions immediately answered by the tutor do not qualify. *Affect acknowledgement* records explicit recognition of feeling or difficulty, excluding generic politeness and praise. These are behavior endpoints, not three assumed independent personality factors.

Five secondary events cover worked explanation, learner choice, epistemic qualification, requests for missing facts, and unsupported ability attribution. All are retained regardless of their results. The pilot showed why a high aggregate agreement rate is insufficient: requesting missing facts had only one majority-coded positive example among 160 responses, while unsupported ability attribution had weak positive agreement. Neither is promoted to a primary endpoint based on subsequent confirmation results.

MiniMax-M3, DeepSeek-V4-Pro, and GLM-5.3 each code every visible tutor response. Coders receive the student conversation, the reference target, and the answer with original line identifiers. They do not receive the generating model name, system-role assignment, repeat number, or partition label. Natural output style can still reveal identity. Positive codes must reference existing nonempty source lines; negative codes have no references. Valid references establish traceability, not semantic truth. We use majority labels when all three coders are structurally valid, and retain individual and leave-one-coder analyses, including fractional means for two-coder ties.

The initial exact-quotation protocol failed on naturally formatted answers, which motivated line references during measurement development. Original failures remain archived. The finalized pilot has 480 valid codes for 160 unchanged tutor responses after two structure/completion failures were retried under identical inputs. Formal structure failures are likewise recorded and retried without selecting by event label or coder agreement. A training coding interruption was recovered using saved valid outputs and the existing two-pass repair allowance. This yielded 3,839 of 3,840 valid codes. One GLM coding request repeatedly exhausted its 16,384-token completion budget without visible output. Before forecast locking or confirmation generation, a disclosed amendment allowed this single missing code a 32,768-token ceiling and at most two attempts, keeping its model, input, rubric, and temperature fixed. It completed on the first attempt using 3,511 completion tokens; this does not identify the cause of the preceding failures. The original and expanded-budget groups were validated separately before their disjoint codes were combined. All 1,280 training answers consequently have three valid codes. The interrupted run, failed attempts, and amendment receipt are retained; the single expanded-budget code is an explicit deviation from the original uniform-budget protocol.

B.4 Prediction Before Confirmation Generation

For each behavior, four nested logistic predictors model the probability of its occurrence: a subject-by-student-state baseline; that baseline plus model identity; an additional model-by-subject block; and an additional model-by-student-state block. New feature blocks are orthogonalized against existing blocks using training data and standardized under source-equal weights. This prevents duplicate main-effect encodings from changing their ridge penalty when interactions are introduced. The intercept is unpenalized. The penalty is selected from $\{0.1, 1, 10, 100\}$ by source-grouped training-only validation. Constant training outcomes use the same smoothed constant across all four predictors.

The fitted models, tuning choices, input hashes, and confirmation probabilities are saved before any confirmation tutor response is generated. The generation runner checks this prediction lock. No confirmation label, answer length, or same-item cross-model score enters a primary predictor. A secondary competing forecast uses only training-source profiles of log word count and marked-line fraction, recomputing those profiles within training validation. Such style competition is descriptive: teaching policy may itself influence length and formatting, so it is not a causal adjustment for style.

For each of the three primary events, the two primary comparisons are the Brier improvement from a default model profile over the situation baseline and from a student-state profile over the model-by-subject baseline. We average losses within source and then equally across the 32 confirmation sources. The six one-sided paired source-level tests use Holm correction. We report absolute Brier losses, source-bootstrap intervals, and all comparison signs. These intervals condition on the fitted training profiles and a source-sampling approximation; they do not describe uncertainty over all models or all educational contexts.

B.5 Repetition and Prompt Plasticity

Repetition analyses report positive, negative, and total event agreement and profile correspondence, retaining source clustering. Constant behavior can produce perfect repeat agreement and is reported as such. On the 16 fully crossed intervention sources, we estimate each model's event-rate difference between an alternative role and the canonical role. We report average shifts alongside model and student-state heterogeneity.

For neutral wording, the fixed equivalence reference is ± 0.10 in event probability. We retain the nominal Bonferroni paired-t intervals across the 30 model–event–neutral-arm comparisons. A nonsignificant difference does not establish equivalence, and constant source differences receive a degeneracy flag. Further synthetic diagnostics during training show undercoverage in a sparse boundary case even with nonzero source variance. Before confirmation generation, we therefore additionally require a bounded-source interval to support any population-equivalence statement. Scaling Hoeffding's bound to source differences in $[-1, 1]$ and allocating error across both tails and all 30 comparisons gives half-width $\sqrt{2\log(2 \cdot 30/0.05)/n}$ (Hoeffding, 1963). At $n = 16$ this is approximately 0.941, too wide to support the chosen tolerance. This conservative sensitivity still assumes independent source observations and does not make the authored bank representative. We can describe observed neutral shifts and nominal intervals, while withholding population equivalence. These disclosed interpretation additions preserve the frozen numerical output and the six primary prediction tests. The explicit ASK and EXPLAIN arms quantify policy displacement rather than better teaching. Finally, answer revelation and reasoning elicitation are analyzed jointly as four possible combinations. Their evidence lines do not identify which action occurred first, and we do not infer action order from them.

B.6 Sensitivity Analyses and Their Timing

During training coding, after measurement development but before confirmation responses exist, we additionally fix eight alternative coding panels: the three-coder mean, each individual coder, each two-coder mean, and the mean after excluding the generator's model family. For each panel, forecasts are refitted using its training labels and locked before confirmation generation. They are evaluated against the corresponding confirmation labels; two-coder ties remain 0.5. This checks dependence on coding choices but cannot remove biases shared by all coders or supply an external ground truth.

Five further forecast sets each remove one generator configuration from both training and the confirmation target, refitting the baselines and tuning within the retained training data. This checks dependence on a single configuration; it changes the target model mixture and does not test prediction of an unseen model. Neither sensitivity family replaces the six primary comparisons.

We also resample the 30 training source families 200 times, preserving the multiplicity of repeated source draws. We retain the selected regularization relative to total loss weight, without repeating hyperparameter selection, and recompute training-style profiles in each resample. The resulting gain quantiles describe training-source sensitivity conditional on the confirmation material and selected regularization. They are separate from the primary confirmation- source intervals. These extensions were specified before confirmation outputs, not preregistered before the entire research project began.

B.7 Mathematics and Content-Error Sensitivity

The protocol also requires a mathematics subset and a check of dependence on obvious content errors. Its implementation is fixed during training coding, before confirmation generation. We evaluate the existing forecasts on all eight mathematics sources, covering 320 canonical responses, without refitting. Separately, MiniMax-M3 and DeepSeek-V4-Pro review all 1,280 canonical responses for checkable content errors using the student input, reference, and original answer lines. They distinguish a clear error, no clear error identified, and uncertainty. Withholding a solution, asking a question, or correctly quoting and correcting an error is not itself an error. Positive error judgments require a category and source-line evidence.

Twelve agent-authored controls accompany the responses, giving 2,584 planned secondary review requests. Each reviewer must match at least five of six controls in each of the clear-error and no-clear-error classes to meet the limited diagnostic rule. If either reviewer fails, screened results remain measurement-limited diagnostics. These controls do not establish correctness on natural responses, even when the rule is met.

We retain the full primary results and report three additional screens: excluding unanimously identified errors; excluding errors identified by either reviewer; and excluding either errors or uncertainty. A flagged response removes its entire five-model source–student-state–repeat slot so that model comparisons remain paired. We report retained coverage and source-equal losses with descriptive source intervals, without new hypothesis tests. These screens condition on generated responses and change the target distribution; they are not causal controls for ability. The content review covers the canonical panel, not all prompt-intervention responses.